

MINI PROJECT REPORT

Automobile Sales Analytics

Power BI Based Automobile Sales Analytics

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1. ABSTRACT

Automobile sales analysis is an important data analytics application that helps organizations understand vehicle sales performance across different models, regions, fuel types and customer financing patterns. This project implements an interactive Automobile Sales Analytics Dashboard using Microsoft Power BI to analyze Tata Motors vehicle sales data and present meaningful business insights.

The Power BI report is built using the Tata Motors automobile sales dataset, containing information related to vehicle models, variants, dealer locations, sales dates, sales units, selling prices, fuel types, body types and financing details. The dashboard provides key performance indicators such as Total Units Sold, Total Revenue, Average Selling Price, Total Models and Average Mileage.

The dashboard consists of three analytical pages: Executive Sales Dashboard, Brand & Model Analysis, and Regional & Customer Analysis. It provides visualizations such as sales trends, bar charts, column charts, donut charts and scatter plots to analyze sales performance from different perspectives. Interactive slicers for Year, Region, Fuel Type, Model and Finance Status allow users to dynamically filter the analysis.

The project converts raw automobile sales data into an interactive and visually understandable business intelligence solution. The final dashboard helps identify high-performing models, regional sales patterns, fuel-type distribution, pricing trends and financing-related sales patterns, supporting data-driven decision making.

1.1 KEY OUTCOMES

- Analyze overall automobile sales performance using important KPIs.
- Identify the top-performing vehicle models based on sales volume and revenue.
- Compare automobile sales across different regions and states.
- Analyze sales distribution across different fuel types and body types.
- Examine the relationship between average selling price and sales volume.

1.2 KEY FEATURES

- KPI cards for Total Units Sold, Total Revenue, Average Selling Price, Total Models and Average Mileage.
- Sales trend analysis to understand changes in vehicle sales over time.
- Model-wise comparison of sales volume, revenue and average selling price.
- Regional and state-wise analysis of automobile sales.
- Fuel type and body type distribution analysis.
- Price versus sales volume analysis using a scatter chart.
- Finance and non-finance sales comparison.

2. INTRODUCTION

The automobile industry generates large amounts of sales data related to vehicle models, sales volume, pricing, regions, fuel types and financing. Analyzing this information manually can make it difficult to identify important sales patterns and compare the performance of different vehicles and regions. A business intelligence dashboard can help convert this raw data into meaningful and easily understandable information.

In this project, automobile sales data is analyzed using Microsoft Power BI. The dashboard provides an interactive view of Tata Motors automobile sales performance and allows users to explore sales trends, model performance, regional distribution, pricing, fuel types and financing patterns.

2.1 PROBLEM STATEMENT

The main problem is to analyze automobile sales data and present the resulting information in a clear, interactive and decision-oriented dashboard. The analysis should help identify high-performing vehicle models, regional sales patterns, pricing trends, fuel-type distribution and financing-related sales information.

2.2 OBJECTIVES

- Analyze financing patterns and bank-wise sales.
- Create meaningful KPI measures for the dashboard.
- Provide interactive filtering using Year, Region, Fuel Type, Model and Finance Status.
- Present the analysis through an interactive and user-friendly Power BI dashboard
- To analyze automobile sales data using Power BI.
- To identify important sales trends and patterns.
- To compare the performance of different vehicle models.
- To analyze regional and state-wise sales performance.
- To understand pricing, fuel-type and financing patterns.
- To provide meaningful insights that can support business decision-making.

2.3 SCOPE

The scope of this project covers automobile sales analysis, KPI reporting, model comparison, regional analysis, pricing analysis, fuel-type analysis and financing analysis using Power BI. It provides interactive visualizations and filtering capabilities that allow users to explore the sales data from different perspectives.

The project focuses on descriptive and analytical insights from the available automobile sales dataset. It does not attempt to predict future vehicle sales or customer behavior; instead, it provides a visual and data-driven view of existing sales performance that can support business analysis and decision-making.

3. DATASET AND DATA MODEL

3.1 DATASET FIELDS

Field	Purpose in Analysis
record_id	Unique record identifier
year	Year associated with the automobile sales record
make	Automobile manufacturer
model	Vehicle model used for model-wise analysis
variant	Specific vehicle variant
dealer_state	State-wise sales analysis
dealer_city	Dealer location analysis
region	Regional sales comparison
sales_date	Used for sales trend analysis
sales_units	Number of vehicles sold
sale_price_inr	Selling price used for revenue and price analysis
msrp_inr	Manufacturer's suggested retail price
fuel_type	Fuel-type sales comparison
engine_cc	Engine capacity information
transmission	Transmission type
body_type	Body-type sales analysis
seats	Seating capacity
mileage_kmpl	Mileage analysis
bank_name	Bank-wise financing analysis
finance_flag	Finance vs non-finance comparison
monthly_emi_inr	Average EMI analysis

3.2 DATASET OVERVIEW

The project uses the tata_cars_100k dataset for automobile sales analysis in Power BI. The dataset contains 100,000 records and 71 original attributes covering vehicle information, sales details, dealer locations, pricing, fuel types and financing information.

For the dashboard development, 21 relevant attributes were selected from the original dataset. These fields were chosen to support sales analysis, model comparison, regional analysis, vehicle characteristics and financing-related analysis.

The selected attributes include record ID, year, make, model, variant, dealer state, dealer city, region, sales date, sales units, selling price, MSRP, fuel type, engine capacity, transmission, body type, seats, mileage, bank name, finance flag and monthly EMI.

3.3 DATA MODEL

The Power BI report contains the main tata_cars_100k table along with a separate DateTable created for time-based analysis. The DateTable contains Date, Year, Month Number, Month Name and Year Month fields.

A relationship is established between tata_cars_100k[sales_date] and DateTable[Date]. The relationship enables time-based analysis such as yearly and monthly sales trends.

The tata_cars_100k table also contains the calculated measures used throughout the dashboard, including Total Units Sold, Total Revenue, Average Selling Price, Total Models and Average Mileage.

3.4 DATA QUALITY

Before creating the dashboard, the dataset was checked for data quality issues. The final working dataset contains 100,000 records and 21 selected attributes. Duplicate record IDs were checked and no duplicate records were removed.

Important numerical and date fields were checked for errors and empty values. Sales units contained valid values, while sales price and sales date fields were also verified. The bank_name field contains blank values because bank information is not applicable to non-financed vehicle sales; these values were retained in the dataset and filtered where required for bank-specific visualizations.

4. TECHNOLOGIES AND TOOLS

Technology	Use
Microsoft Power BI	Dashboard development, visual analytics and interactive filtering.
Power Query	Data preparation and transformation where required.
DAX	Creation of analytical measures and business KPIs.
Customer Segmentation Output	Primary analytical dataset used by the report.
GitHub	Project repository and versioned project resources.

4.1 POWER BI DASHBOARD DESIGN

The Power BI dashboard is organized into three analytical pages: Executive Sales Dashboard, Brand & Model Analysis, and Regional & Customer Analysis. The dashboard uses KPI cards, charts and interactive slicers to provide a comprehensive view of automobile sales performance. A consistent visual design with a dark blue header, light background and white visual areas is used across the dashboard.

4.2 DASHBOARD COMPONENTS

- Total Units Sold – total number of vehicles sold.
- Total Revenue – total revenue generated from vehicle sales.
- Average Selling Price – average selling price of the vehicles.
- Total Models – total number of distinct vehicle models.
- Average Mileage – average mileage of the vehicles.
- Sales Trend – line chart showing sales volume over time.
- Sales by Region – comparison of vehicle sales across regions.
- Top 10 Models by Sales – identifies the highest-selling vehicle models.
- Sales by Fuel Type – donut chart showing sales distribution across fuel types.
- Sales by Model – comparison of total vehicle sales by model.
- Revenue by Model – comparison of revenue generated by different models.
- Average Selling Price by Model – comparison of average prices across models.
- Sales by State – state-wise comparison of vehicle sales.
- Finance vs Non-Finance Sales – comparison of financed and non-financed vehicle sales.
- Sales by Bank – comparison of vehicle sales associated with different banks.
- Average EMI by Region – comparison of average monthly EMI across regions.
- Sales by Body Type – distribution of sales across different vehicle body types.
- Interactive Slicers – Year, Region, Fuel Type, Model and Finance Status filters for interactive analysis

5. DATA PREPARATION AND ANALYTICAL MEASURES

The automobile sales dataset contains the fields required to analyze major sales dimensions such as sales volume, revenue, pricing, vehicle models, regional performance and financing. These dimensions provide a practical basis for understanding automobile sales performance and identifying important business patterns.

5.1 SALES VOLUME

Sales volume represents the number of vehicles sold. In the dashboard, Total Units Sold is used to compare sales performance across different models, regions, fuel types and other categories.

5.2 SALES REVENUE

Sales revenue represents the revenue generated from vehicle sales. It is calculated using the number of vehicles sold and their selling prices and is used to compare the financial performance of different vehicle models and regions.

5.3 SELLING PRICE

Selling price represents the actual price at which vehicles are sold. Average Selling Price is used in the dashboard to compare pricing levels across different vehicle models and to study the relationship between vehicle price and sales volume.

5.4 DASHBOARD MEASURES

Measure	Dashboard Purpose
Total Units Sold	Shows the overall number of vehicles sold.
Total Revenue	Shows the total revenue generated from vehicle sales.
Average Selling Price	Shows the average selling price of vehicles.
Total Models	Shows the number of distinct vehicle models.
Average Mileage	Shows the average mileage of the vehicles.
Average EMI	Shows the average monthly EMI for financing analysis.

These measures are displayed through KPI cards and dashboard visuals so that users can immediately understand the overall sales performance before examining individual models, regions and other categories.

6. DATA PREPARATION AND ANALYTICAL MEASURES

The dashboard uses multiple visual forms to analyze automobile sales from different perspectives. Sales volume, revenue, pricing, model performance, regional distribution, fuel type and financing patterns are examined together to provide a comprehensive view of automobile sales performance.

6.1 SALES TREND

The line chart shows the total number of vehicles sold over time. This provides an overview of changes in sales volume across different months and years and helps identify variations in automobile sales over the selected period.

6.2 SALES BY REGION

The sales by region chart compares the total number of vehicles sold across different regions. This helps identify regions with higher or lower sales performance and provides a geographical view of automobile demand.

6.3 TOP 10 MODELS BY SALES

The Top 10 Models by Sales chart compares vehicle models based on total units sold. It provides a quick view of the highest-performing models and helps identify models with greater sales volume.

6.4 PRICE VS SALES VOLUME

The scatter chart compares Average Selling Price with Total Units Sold for different vehicle models. This helps examine whether differences in vehicle pricing are associated with differences in sales volume.

6.5 SALES BY FUEL TYPE

The donut chart presents the distribution of total vehicle sales across different fuel types. It provides an overview of the contribution of each fuel category to overall automobile sales.

6.6 REGIONAL AND STATE-WISE SALES

The regional and state-wise charts compare automobile sales across geographical locations. These visualizations help identify areas with relatively higher sales and provide a more detailed understanding of regional sales patterns.

6.7 FINANCE AND BANK-WISE ANALYSIS

The finance analysis compares financed and non-financed vehicle sales and examines sales associated with different banks. The analysis also includes average EMI by region to provide an overview of financing patterns across geographical areas.

6.8 BODY TYPE ANALYSIS

The body type chart shows the distribution of vehicle sales across different body types. This helps understand which types of vehicles contribute more to the overall sales volume.

7. POWER BI DASHBOARD

The final Power BI dashboard brings together the analytical components into an interactive three-page report. The report provides KPI cards, charts, slicers and interactive filtering to analyze automobile sales from different perspectives.

The dashboard is organized into three pages: Executive Sales Dashboard, Brand & Model Analysis, and Regional & Customer Analysis. The pages allow users to move from overall sales performance to detailed model, regional and financing analysis.

The Executive Sales Dashboard presents the overall sales performance using KPI cards for Total Units Sold, Total Revenue, Average Selling Price, Total Models and Average Mileage. It also includes sales trends, regional sales, top-performing models and fuel-type distribution.



The Brand & Model Analysis page focuses on comparing vehicle models using total sales, revenue and average selling price. The Price vs Sales Volume scatter chart provides a visual comparison between vehicle pricing and sales volume.

The Regional & Customer Analysis page presents regional and state-wise sales, finance versus non-finance sales, bank-wise sales, average EMI by region and body-type sales distribution.

7.1 INTERACTIVE FEATURES

- Year slicer for filtering the dashboard by year.
- Region slicer for analyzing sales across different regions.
- Fuel Type slicer for comparing different fuel categories.
- Model slicer for examining individual vehicle models.
- Finance Status slicer for comparing financed and non-financed sales.
- Cross-filtering between dashboard visuals.
- KPI cards that update based on the selected filters.
- Interactive comparison of sales, revenue, pricing and regional performance.

8.AUTOMOBILE SALES INTERPETATION

The dashboard provides several important business interpretations related to vehicle sales, model performance, regional distribution, pricing, fuel types and financing. These interpretations help understand the major patterns present in the automobile sales data and provide useful information for business decision-making.

Analysis Area	Observed Interpretation	Business Meaning
Model Performance	Some vehicle models contribute higher sales volume than others.	High-performing models can be given greater attention in sales and marketing strategies.
Regional Performance	Sales volumes vary across different regions and states.	Regional differences can help identify strong markets and areas requiring additional attention.
Pricing	Average selling prices differ across vehicle models.	Pricing information can help understand the relationship between vehicle price and sales performance.
Fuel Type	Vehicle sales are distributed across different fuel types.	Fuel-type preferences provide insight into the composition of automobile demand.
Financing	Vehicle sales include both financed and non-financed purchases.	Financing patterns can help understand the role of vehicle financing in sales.
Body Type	Sales vary across different vehicle body types.	Body-type preferences can help identify categories with stronger demand.

8.1 MODEL PERFORMANCE INSIGHT

The model-wise analysis helps identify vehicles with higher sales volume and revenue. The Top 10 Models by Sales visualization provides a quick comparison of the best-performing models and can help identify models that contribute significantly to overall sales.

8.2 REGIONAL SALES INSIGHT

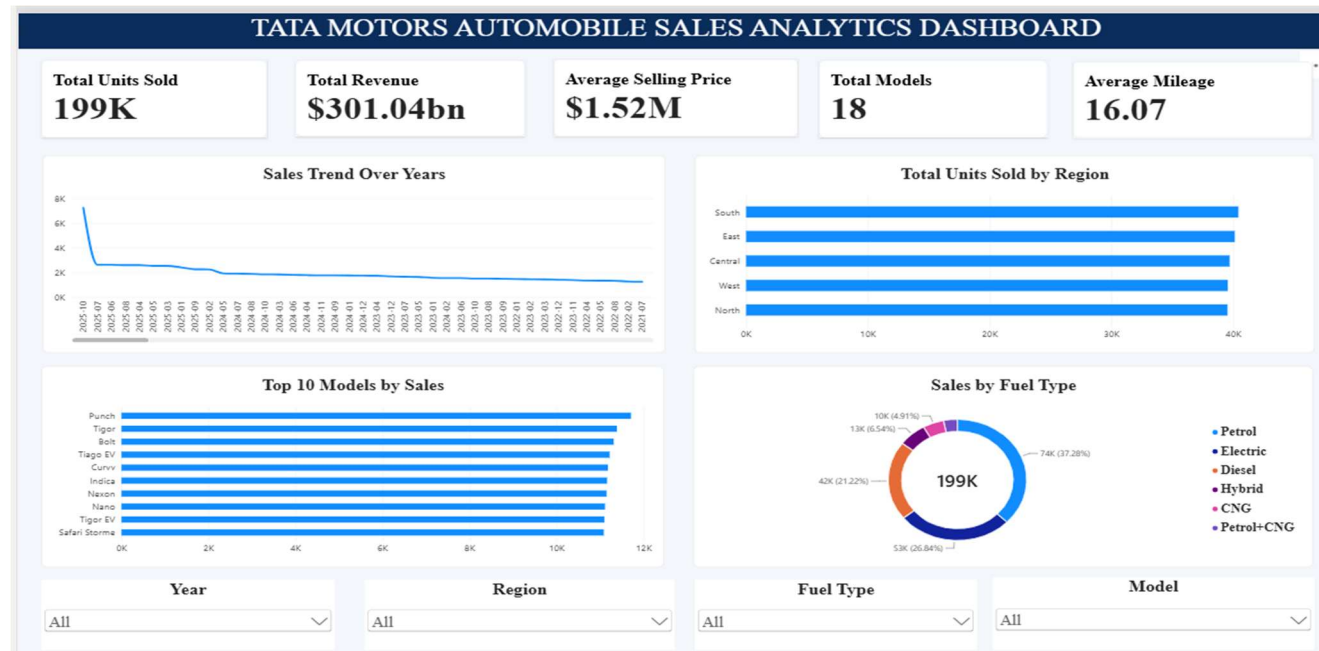
The regional and state-wise analysis shows that automobile sales are distributed differently across geographical areas. Comparing regions and states helps identify areas with relatively stronger sales performance and provides useful information for regional business analysis.

8.3 PRICING INSIGHT

The Price vs Sales Volume scatter chart provides a comparison between average selling price and total units sold for different models. This visualization helps examine how vehicle pricing varies in relation to sales volume.

9. RESULTS AND BUSINESS INSIGHTS

The Power BI report provides a consolidated view of Tata Motors automobile sales performance. The analysis presents important results related to sales volume, revenue, vehicle models, regional performance, pricing, fuel types and financing patterns. The dashboard visuals help identify major sales trends and provide useful business insights from the available automobile sales data.



9.1 MAJOR FINDINGS

- The dashboard provides an overall view of automobile sales through Total Units Sold, Total Revenue, Average Selling Price, Total Models and Average Mileage.
- Sales performance varies across vehicle models, allowing high-performing and relatively lower-performing models to be identified.
- Regional & state-wise visualizations show differences in sales distribution across geographical areas.
- Top 10 Models by Sales visualization highlights the models contributing the highest sales volume.
- Revenue varies across different models, providing an understanding of the financial contribution of individual vehicle models.
- Price vs Sales Volume analysis helps compare vehicle pricing with their corresponding sales volume.
- Fuel-type and body-type visualizations show how automobile sales are distributed across different vehicle categories.

9.2 BUSINESS RECOMMENDATIONS

- Focus marketing and sales efforts on high-performing vehicle models.
- Analyze lower-performing models to identify opportunities for improving their sales performance.
- Use regional and state-wise sales patterns to support location-specific sales and marketing strategies.
- Consider pricing and sales-volume relationships when evaluating vehicle pricing strategies.
- Monitor fuel-type and body-type sales patterns to understand changing demand across vehicle categories.

10. CONCLUSION AND FUTURE SCOPE

10.1 CONCLUSION

The Automobile Sales Analytics Dashboard project demonstrates how automobile sales data can be transformed into useful business information using Microsoft Power BI. The Tata Motors sales dataset provides the foundation for analyzing sales volume, revenue, vehicle models, pricing, regional performance, fuel types and financing patterns.

The dashboard combines KPI cards, sales trends, model comparisons, regional analysis, pricing analysis, fuel-type distribution and financing-related visualizations. Interactive slicers allow users to explore the data based on Year, Region, Fuel Type, Model and Finance Status. This combination makes the analysis easier to understand and supports data-driven decision-making.

The project highlights that automobile sales performance can be studied from multiple perspectives rather than relying on a single sales metric. Model-wise, regional, pricing, fuel-type and financing analyses provide a broader understanding of sales performance and can help identify areas requiring further attention.

10.2 FUTURE SCOPE

- Add sales forecasting using machine learning or time-series models.
- Develop predictive models for future vehicle demand.
- Integrate real-time or regularly updated automobile sales data.
- Add customer-level demographic information for more detailed customer analysis.
- Include profit and cost analysis in addition to revenue.
- Add dealer-level performance analysis and comparison.
- Develop automated data refresh mechanisms for regularly updated dashboards.
- Deploy the dashboard through Power BI Service for organizational access.

10.3 LIMITATIONS

- The analysis is limited to the information available in the Tata Motors automobile sales dataset.
- The dashboard provides descriptive and analytical insights but does not directly predict future sales.
- Customer demographic information is not available in the selected dataset, limiting detailed customer-level analysis.
- Blank bank values are present because financing information is not applicable to some sales records.
- The accuracy of the analysis depends on the quality and completeness of the underlying dataset.
- Business decisions based on the dashboard should be validated using additional business information and real-world market conditions.

11. PROJECT SUMMARY

The Automobile Sales Analytics Dashboard project focuses on analyzing Tata Motors automobile sales data using Microsoft Power BI. The project transforms a large automobile sales dataset into an interactive dashboard that presents important information related to sales volume, revenue, vehicle models, pricing, regions, fuel types and financing patterns.

The dataset contains 100,000 records and 71 original attributes. For dashboard development, 21 relevant fields were selected to support sales, vehicle, geographical and financing analysis. The data was prepared using Power Query, and DAX measures were created to calculate important KPIs such as Total Units Sold, Total Revenue, Average Selling Price, Total Models, Average Mileage and Average EMI.

The final Power BI report is organized into three pages: Executive Sales Dashboard, Brand & Model Analysis, and Regional & Customer Analysis. Different visualization techniques, including line charts, bar charts, column charts, donut charts and scatter charts, are used to present the analysis in an easy-to-understand manner.

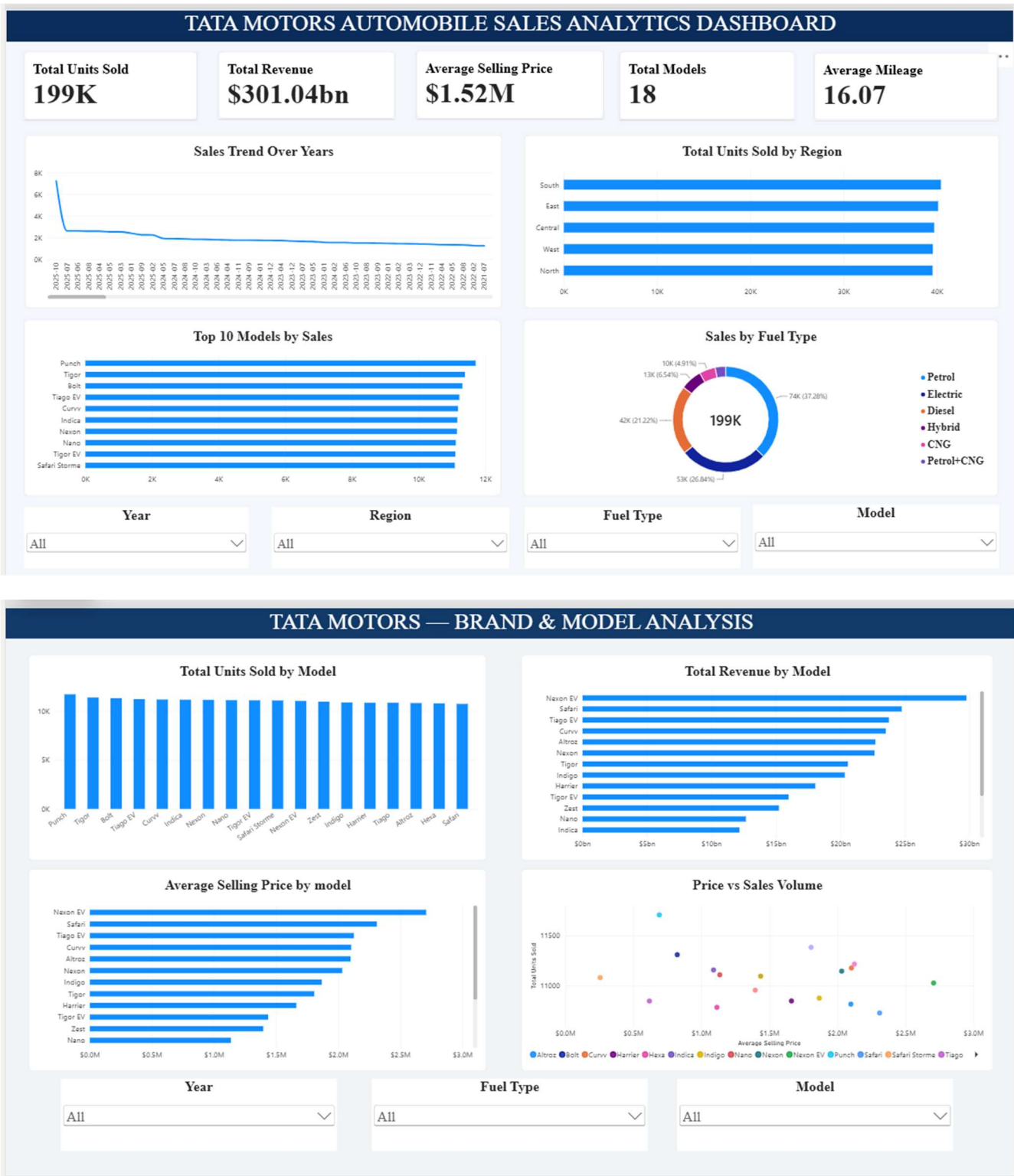
Item	Details
Project	Automobile Sales Analytics Dashboard
Primary Platform	Microsoft Power BI
Analytical Dataset	Tata Motors Automobile Sales Dataset
Dataset Size	100,000 records and 71 original attributes
Selected Fields	21 relevant sales, vehicle, regional and financing fields
KPIs	Total Units Sold, Total Revenue, Average Selling Price, Total Models, Average Mileage
Main Visuals	Line chart, clustered bar chart, clustered column chart, donut chart, scatter chart
Dashboard Pages	Executive Sales Dashboard, Brand & Model Analysis, Regional & Customer Analysis
Filters	Year, Region, Fuel Type, Model and Finance Status slicers
Analytical Measures	Total Units Sold, Total Revenue, Average Selling Price, Total Models, Average Mileage, Average EMI
Main Business Focus	Sales performance, model comparison, regional analysis, pricing and financing patterns

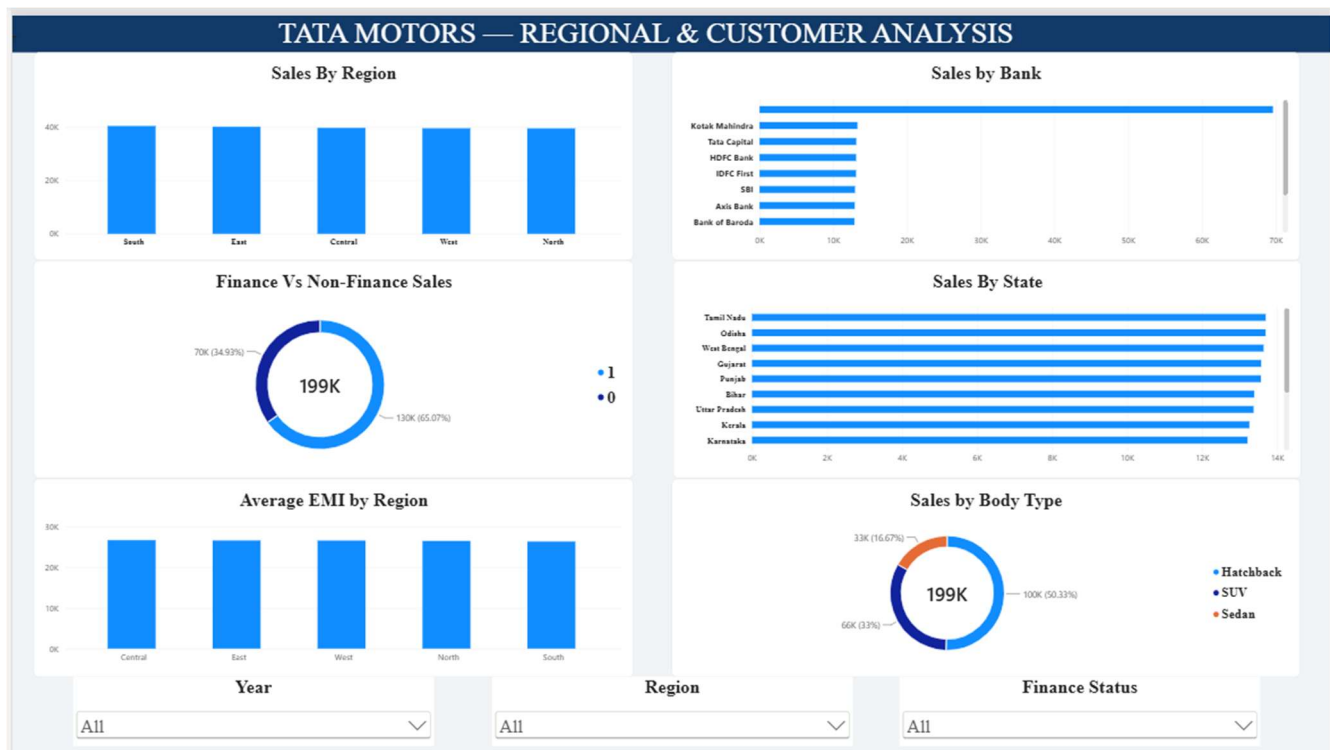


12.1 GITHUB REPOSITORY

Repo Link: https://github.com/MalluriHarshith/Tata_Automobile_Sales/tree/main

12.2 FULL DASHBOARD OVERVIEW





REFERENCES

- Microsoft Power BI – Data visualization and business intelligence platform.
- Microsoft Power Query – Data preparation and transformation tool used for data cleaning and preprocessing.
- Microsoft DAX – Formula language used for creating analytical measures and KPIs in Power BI.
- Tata Motors – Official website and automobile product information.
- Tata Motors Automobile Sales Dataset – Dataset used for automobile sales analysis and dashboard development.
- GitHub – Project repository used for storing and managing project resources.